

Kalp Shah

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PROFESSIONAL SUMMARY

Software Engineer focused on the intersection of embedded Linux IoT platforms and cloud infrastructure. Experienced in designing, scaling, and automating real-time systems that monitor, control, and optimize power electronics and battery energy storage systems (BESS). Proven expertise across the complete edge-to-cloud pipeline, from low-latency hardware protocol integration (Modbus, CANbus, MQTT) to high-throughput AWS telemetry, Infrastructure as Code, and resilient edge runtimes.

EXPERIENCE

Energy Toolbase

Calgary, AB

Intermediate Software Engineer

June 2023 – Present

- Architect and develop embedded Linux IoT platform software and cloud infrastructure to monitor, automate, and optimize power electronics and battery energy storage systems (BESS) in real time.
- Drive technical design and platform architecture for edge systems, collaborating directly with product stakeholders to translate requirements into resilient, production-ready services.
- Build scalable web APIs and edge-to-cloud transport mechanisms using Python, FastAPI, and Go, interfacing with external hardware devices via Modbus (RTU/TCP), CANbus, Digital I/O, and MQTT.
- Expand telemetry integration with AWS cloud services to ensure high-throughput ingestion pipelines, automated alerting, and low-latency bidirectional device control.
- Streamline deployment consistency across edge and cloud environments using Infrastructure as Code (IaC), SaltStack, Docker, Kubernetes, and CircleCI.
- Ensure platform reliability and security through comprehensive unit/integration testing (TDD/BDD), secure edge-cloud communication protocols, and active production on-call rotation.
- Facilitate code reviews, evaluate emerging technologies, and provide peer mentorship to elevate team-wide engineering standards and architectural best practices.

Software Engineer

Jan. 2021 – June 2023

- Developed and maintained core edge device software services, hardware communication interfaces, and automated test suites for commercial and industrial energy storage deployments.
- Implemented data parsing, state-machine control logic, and local fault mitigation for power conversion systems, improving edge reliability under diverse grid conditions.

Celestica

Newmarket, ON

Test & Automation Developer

May 2018 – Sept. 2019

- Collaborated within a team of 12 senior engineers to design and develop automated test solutions for a tier-1 telecommunications client in the United States.
- Engineered automated test scripts and internal tooling to streamline manufacturing verification processes and improve testing cycle efficiency.
- Built regression test suites and data logging utilities to identify hardware defects early in the production lifecycle.

PROJECTS

Second Brain: Advanced Hybrid RAG Knowledge System | *FastAPI, Qdrant, BM25, ONNX, Ollama, Docker, Python*

- Architected an offline-first Personal Knowledge Management RAG system combining dense vector search (Qdrant, BGE) and sparse lexical search (BM25) via Reciprocal Rank Fusion (RRF, $k = 60$) to eliminate keyword blindness.
- Engineered hierarchical parent-child chunking with contextual breadcrumbs and cross-encoder reranking (FlashRank ONNX), improving retrieval precision by 35% and feeding complete contextual sections to local LLMs (Ollama).
- Implemented automated RAG Triad observability (Context Relevance, Faithfulness, Answer Relevance) and built an interactive dark-mode dashboard featuring SSE token streaming and force-directed knowledge graph visualization.

Personal Cloud Platform & Infrastructure | *Docker Compose, Cloudflare Tunnels, MySQL, Bash, Python*

- Architected and deployed self-hosted multi-service production infrastructure (kalp.dev) featuring zero-trust ingress via Cloudflare Tunnels.
- Developed automated disaster recovery pipelines with systemd timers, Bash, and offsite storage, validated by Python integration test suites.

EDUCATION

Ryerson University

Toronto, ON

Bachelor of Engineering (B.Eng.) in Computer Software Engineering

TECHNICAL SKILLS

Languages: Python, Go, C/C++, SQL, Bash, JavaScript

IoT & Hardware Protocols: Modbus (RTU/TCP), CANbus, MQTT, Digital I/O, Embedded Linux, Power Electronics, EMS

Cloud & DevOps: AWS, Docker, Kubernetes, SaltStack, CircleCI, Linux, Infrastructure as Code (IaC), Git, CI/CD

Frameworks & Architecture: FastAPI, Hybrid RAG, Qdrant (Vector DB), REST APIs, Microservices, TDD/BDD, System Design